Prediction for Economists

Prediction as an objective

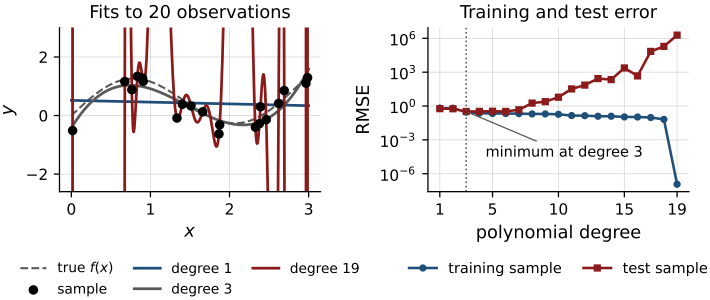
- A rule: observed characteristics in, an unobserved outcome out.
- Estimated on one sample, judged on another.
- The criterion is error on the second, not the value of any coefficient.
- Bias in the coefficients is acceptable when it lowers that error.

Everything that follows is about estimating that error honestly.

Outline

1. Overfitting and the trade-off
2. Estimating out-of-sample error
3. Leakage
4. Measures of accuracy
5. Constructing regressors
6. Choosing an estimator
7. Putting it together

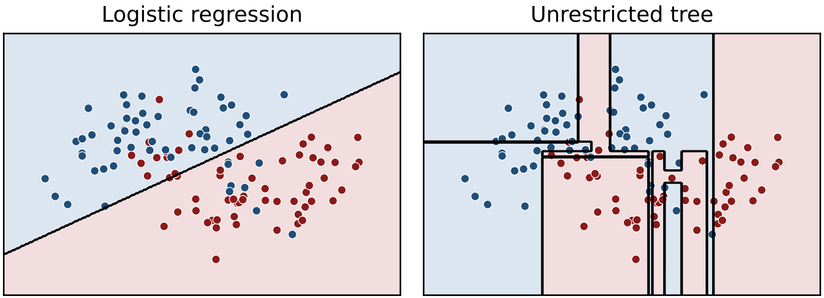
Overfitting



- In-sample fit: better at every degree.
- 20 observations, degree 19: every point exactly.
- Error in an independent sample: lowest at a low degree.

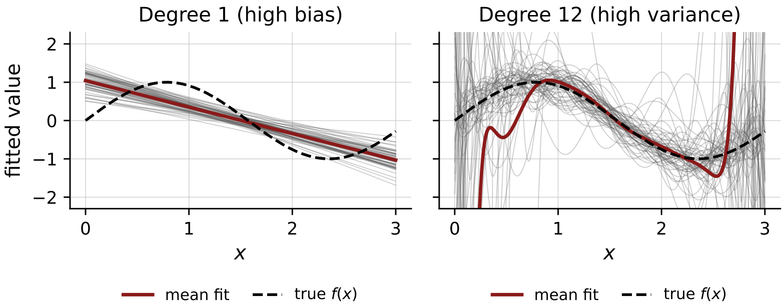
Fit in the estimation sample is not evidence of accuracy elsewhere.

Overfitting with a binary outcome



- Overfitting is not a property of polynomials or of continuous outcomes.
- Left unrestricted, the tree keeps splitting until each region is pure.
- It ends up isolating single observations, and misclassifies new ones.

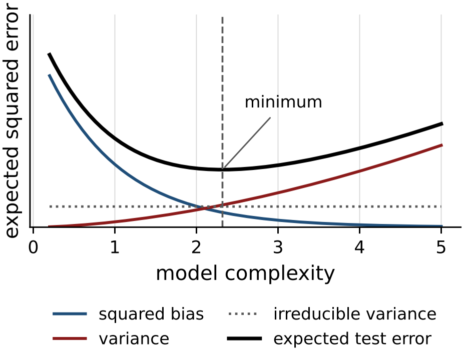
Bias and variance



- Each grey line: one independent sample of 25 observations.
- Restrictive fit: stable across samples, wrong on average.
- Flexible fit: right on average, far off in any one sample.

Two distinct errors, and less of one buys more of the other.

The bias–variance trade-off



- Squared bias, variance, noise.
- Restrictive: bias dominates.
- Flexible: variance dominates.
- The noise is irreducible.
- One point on the curve is best.

Which point, though? The curve is drawn here because the truth is known.

Outline

Holding observations back

Training	Validation	Test

- Out-of-sample error is unobserved. Hide observations and imitate it.
- Training: estimate the coefficients of one given specification.
- Validation: measure error there to *choose* — which regressors, which estimator, how flexible.
- Test: measure error once more, to *report* what the chosen rule will do.

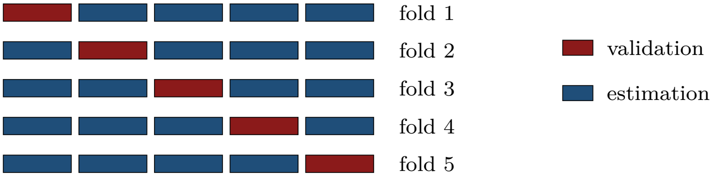
Both measure error; they answer different questions.

Why the test sample is separate

- Compare twenty specifications on the validation sample. One wins.
- It won partly on merit, partly because that sample's noise suited it.
- So the winner's validation error is better than its true accuracy.
- The more specifications compared, the larger that gap.

The test sample took part in no comparison, so its error is not flattered by having been chosen on. That is the number to report.

Cross-validation



- One validation block wastes those observations, and is one arbitrary draw.
- Instead, cut the training data into k folds of equal size.
- Each fold takes the validation role in turn; the other $k-1$ estimate.
- Average the k errors: one number per specification, using all the data.

The validation role rotates inside the training data; nothing is permanently set aside. The test sample stays outside the rotation, still unused.

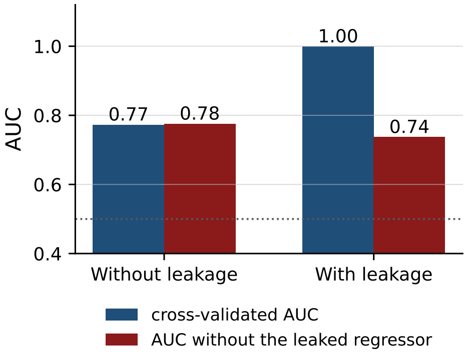
Outline

What leakage is

- Leakage: the fit uses information that will not be available when the rule is actually applied.
- The estimate of accuracy is then an estimate of a task that is easier than the real one.
- It shows up as a good cross-validated error and a disappointing one in use.

Nothing in the procedure complains. The fold was held out, so the arithmetic is correct; it is the data that were contaminated.

Leakage in a cross-validated fit



- Here one regressor was built from the outcome itself.
- Cross-validation reports an almost perfect fit.
- Remove that regressor and the fit is ordinary.

The held-out fold was held out, but its outcome had already travelled into the regressors.

Two kinds of leakage

- Wrong date: a regressor recorded after the outcome, or unknown at the moment the prediction would be made.
- Wrong sample: a quantity computed on the whole data and then used inside the folds.
- The second is the common one: standardization, imputation, category means, the choice of which regressors to keep.

Both make the cross-validated error too low, and the error in use a surprise.

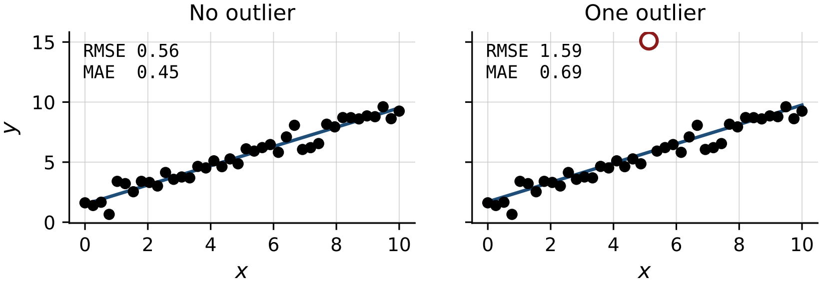
Keeping the estimate honest

- Split first, transform afterwards. Never the reverse.
- Means, standard deviations and imputed values are estimated on the training fold, then applied to the held-out one.
- A regressor enters only if it is known before the outcome is realized.
- The test sample is opened once, at the end.

In code, this is why the transformations and the estimator are fitted together, inside the fold, rather than beforehand.

Outline

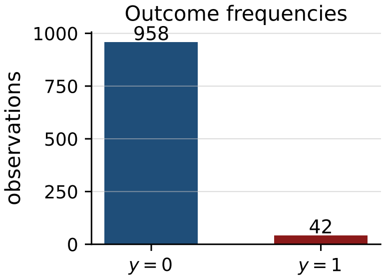
Measuring accuracy: continuous outcomes

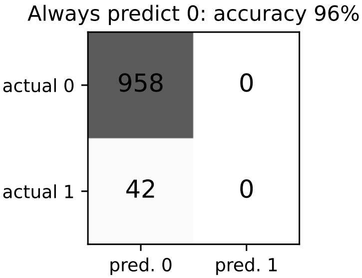


- Root mean squared error: large errors count for more.
- Mean absolute error: every unit of error counts the same.
- A few extreme observations move the first, not the second.

Fix the measure before the estimation, by the cost of an error.

Measuring accuracy: binary outcomes





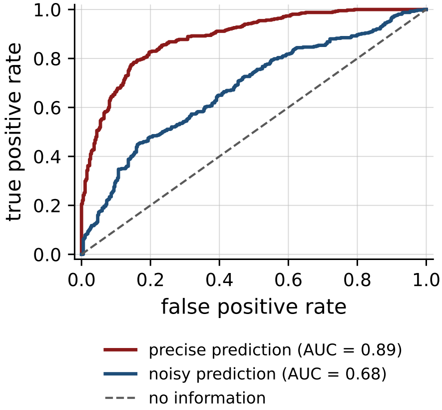
- Rare ones: the share of correct classifications says little.
- Zero everywhere: 96% correct, nothing identified.
- Predicted against actual, as a table: the useful summary.

Thresholds, precision, and recall

- Estimators return a probability. A classification needs a threshold.
- Precision: of those predicted one, the share that are one.
- Recall: of those that are one, the share predicted one.
- A higher threshold: more precision, less recall.

The threshold is a decision about the cost of the two errors, not an estimation problem.

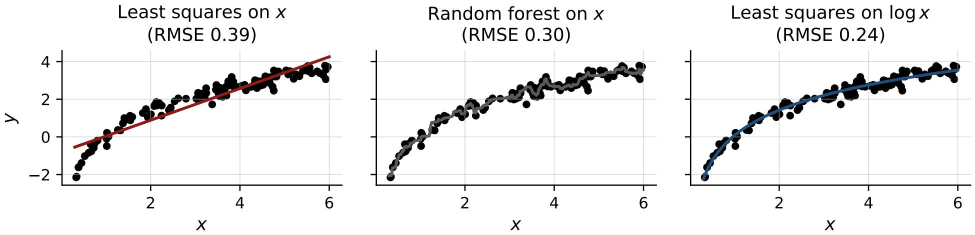
The ROC curve



- Both rates, at every threshold.
- Its area: the quality of the ordering.
- No one threshold involved.
- The diagonal: no information.

Outline

Lesson 1: the regressors matter more than the estimator



- Same 120 observations and same measure in all three panels.
- Changing the estimator, in the middle, removes part of the error.
- Changing the regressor removes nearly all of it: 0.24, against a noise floor of 0.25.

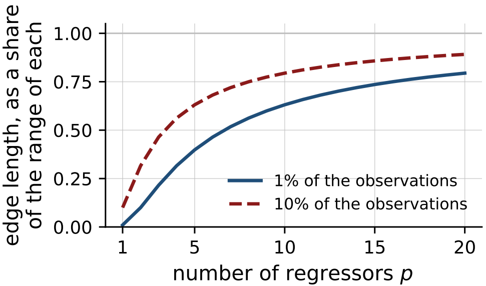
Domingos (2012): the features used are the single most important factor in whether a prediction succeeds, and where most of the work goes.

Lesson 1, applied: what the estimator cannot build for itself

- A linear model fits curvature and interactions only where you supply them.
- A tree builds both itself, but spends observations on every split it needs.
- Monotone transformations ($\log x$, $\sqrt{x}$, ranks) change a linear fit entirely, and leave a tree's splits unchanged.
- Ratios and differences help both: a tree approximates a/b only with many splits, and never exactly.

A date is the clearest case. As a number it is close to useless; as month and day of the week it is often where the signal is.

Lesson 2: every regressor costs data



- Observations spread over the unit cube in p dimensions.
- A neighbourhood holding a fraction r of them needs edge $r^{1/p}$.
- At $p = 10$, holding 1% takes 63% of the range of every regressor.

A neighbourhood that wide is not local, and an estimator that learns from neighbours has nothing left to learn from. This is what an irrelevant regressor costs.

Hastie, Tibshirani and Friedman, sec. 2.5.

Lesson 3: bet on sparsity

- If many regressors each matter a little, no procedure predicts well: the data cannot support that many estimated effects.
- If a few matter a lot, penalized methods find them.
- So act as though few matter. Build the candidates, let the penalty select.
- The bet is not always right. It is that there is nothing better to bet on.

The selected set is not a finding: on another sample a different set appears, predicting about as well.

Hastie, Tibshirani and Friedman, sec. 16.2.2; Mullainathan and Spiess (2017) on the instability of the selection.

Outline

Three decisions

- Which family: penalized linear regression, or something tree-based.
- How flexible within that family: the penalty, the depth, the number of trees.
- When to stop: a gain smaller than the variation across folds is not a gain.

All three are settled by the same number, the cross-validated error, computed on identical folds with identical regressors. Least squares is the first entry, as the benchmark the rest have to beat.

The first decision: which family

Restrictive fits win	**Flexible fits win**
Few observations per regressor	Many observations
Weak signal, much noise	Strong signal
Smooth, additive relations	Thresholds and interactions
Extrapolation is required	Prediction inside the data

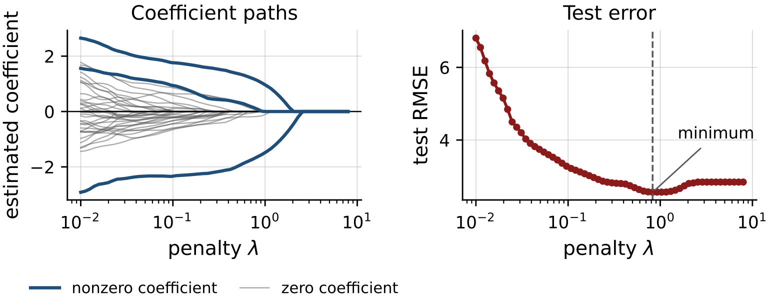
Most outcomes in economics sit on the left: many regressors, modest signal. Each row is a conjecture, and the folds settle it.

The second decision: what each estimator asks you to choose

- Least squares: nothing.
- Ridge and LASSO: one penalty. LASSO also sets coefficients to zero.
- Tree: the depth, or the smallest region allowed.
- Random forest: the number of trees and the regressors per split. Little turns on either.
- Boosting: the rate, the depth, and the number of trees. Much turns on all three.

Each estimator has its counterpart for a binary outcome, with the same choices to make.

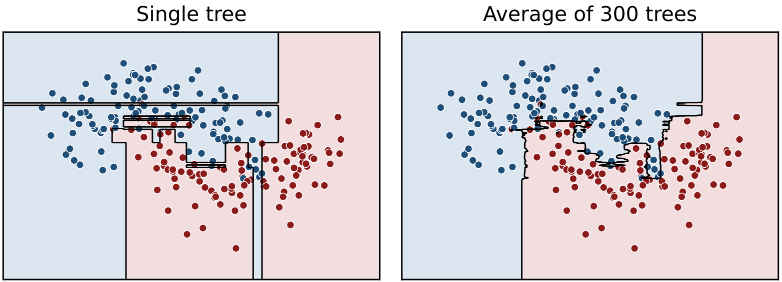
Penalized regression



- 70 observations, 40 regressors, 3 with nonzero coefficients.
- The penalty shrinks the coefficients, and sets the irrelevant to zero.
- Lowest test error at a strictly positive penalty.

The penalty is one more choice, and the folds make it.

Trees and random forests



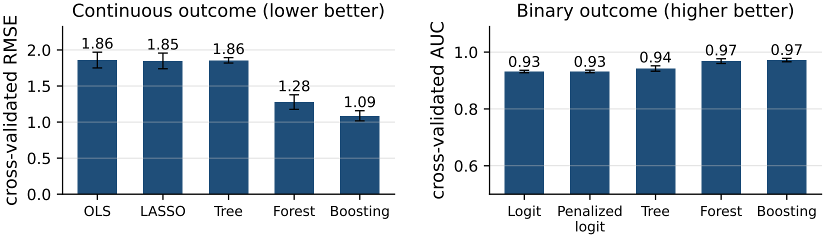
- One tree: the boundary is a feature of the particular sample.
- A forest: many trees on resampled data, averaged.
- The flexibility kept, most of the variance gone.

Boosting

- Trees in sequence, not in parallel.
- Each one fitted to the residuals of the current model.
- Each entering with a small weight, so progress is gradual.
- More parameters than a forest, and it overfits when they are careless.

A forest asks fewer questions of you, and is the better place to start.

Comparing estimators



- Same regressors, same folds, same measure, for every estimator.
- Bars: the average over the five folds. Whiskers: one standard error.
- Flexible fits win here; on data close to linear the ordering reverses.

The third decision, read off the whiskers: on the left boosting is genuinely ahead of the forest, on the right the two are within one standard error and the forest is the simpler rule.

Outline

The workflow

1. Split → 2. Construct regressors → 3. Estimate → 4. Cross-validate → 5. Select → 6. Evaluate

- Was every transformation fitted inside the fold?
- Could any regressor contain the outcome?
- Same regressors and same folds across every specification?

Step 6 happens once. Anything learned there is not a reason to return to step 2.

Summary

- Accuracy concerns observations not used in the estimation.
- In-sample fit does not measure it; bias against variance is the trade-off.
- Cross-validation estimates it. Leakage quietly destroys the estimate.
- Fix the measure of accuracy in advance, by the cost of an error.
- Regressors first, estimator second.

Appendix

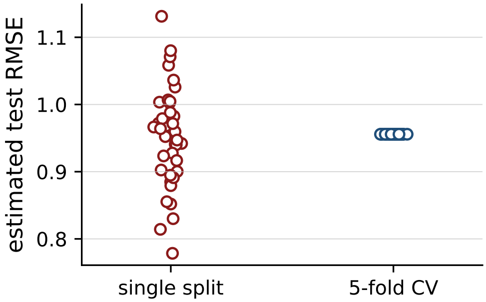
The decomposition of expected prediction error

For a new observation at x_0, with $y = f(x_0) + \varepsilon$ and $\hat{f}$ estimated on a random sample,

$$\mathbb{E}\Big[(y - \hat{f}(x_0))^2\Big] = \big(\mathbb{E}[\hat{f}(x_0)] - f(x_0)\big)^2 + \mathrm{Var}\big(\hat{f}(x_0)\big) + \mathrm{Var}(\varepsilon).$$

- The expectation is taken over samples, not over observations.
- The first two terms can be traded against one another.
- The third cannot be reduced by any estimator.
- Penalization and averaging lower the second at some cost in the first.

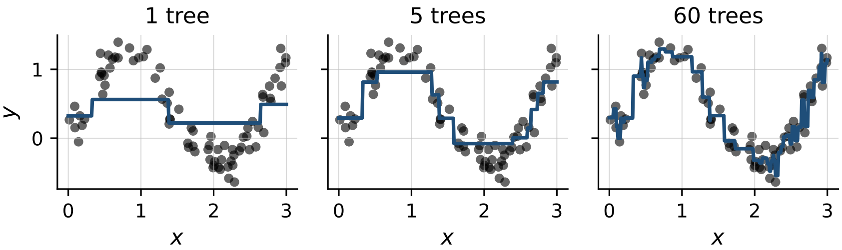
Sampling variability of a single split



- Each point: a new partition.
- One split varies a great deal.
- The average over folds is stable.
- Small samples: it matters most.

With a binary outcome, keep the share of ones equal across folds.

Boosting: successive fits



- A single tree of depth two is a coarse step function.
- Successive trees fitted to the residuals accumulate into an accurate fit.
- Continued far enough, the procedure begins to fit the noise.
- The number of trees is itself chosen by cross-validation.

Categorical variables: the encoding is the decision

- Few levels: one indicator each, and nothing further is needed.
- Many levels: as many regressors as levels, most of them nearly empty, and each estimated on a handful of observations.
- Group the rare levels together, or replace a level by its frequency.
- Replacing a level by its mean outcome predicts well and leaks: the outcome enters the regressor. Compute it within the fold or not at all.

Missing values carry information

- Values are rarely missing at random: no credit history, a firm too young to report, a question refused.
- Impute the value, and add an indicator for having imputed it.
- The indicator is often the better predictor of the two.
- Both the imputation and the indicator are built inside the fold.

Dropping the incomplete observations throws that information away, and leaves a sample selected on the regressors.

Further measures for binary outcomes

- When ones are rare, the precision–recall curve is more informative.
- ROC can look favourable when few of the predicted ones are correct.
- The F_1 measure combines precision and recall in one number.
- Log-loss penalizes confident predictions that turn out to be wrong.
- Calibration: of those given probability 0.7, about 70% should be ones.

References

- Mullainathan and Spiess (2017), *Journal of Economic Perspectives* 31(2).
- Athey and Imbens (2019), *Annual Review of Economics* 11, 685–725.
- Domingos (2012), *Communications of the ACM* 55(10), 78–87.
- James, Witten, Hastie and Tibshirani, *Introduction to Statistical Learning*.
- Hastie, Tibshirani and Friedman, *The Elements of Statistical Learning*.